PATENT SPECIFICATION

Method for Quantifying Anomalies Using Multi-Spectrum Tensor Decomposition Across Independent Mathematical Law Domains

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1. TECHNICAL FIELD

The present invention relates to the field of anomaly detection and machine learning, and more particularly to a model-independent framework for detecting and characterising anomalies by measuring deviations across multiple independent mathematical representations of data.

2. BACKGROUND OF THE INVENTION

2.1. Technical Problem

Anomaly detection—the identification of observations that deviate significantly from expected patterns—is a foundational problem in data analysis with applications across finance, cybersecurity, industrial monitoring, medical diagnosis, and scientific research. Existing approaches suffer from several structural limitations:

(a) **Single-perspective detection:** Standard anomaly detectors (Isolation Forest, Local Outlier Factor, One-Class SVM, autoencoders) produce a single anomaly score. They identify *that* an observation is anomalous but not *which mathematical aspect* makes it anomalous. This limits interpretability and actionability.

(b) **Detector-dependent representations:** Each detector imposes its own implicit representation of "normality." An observation that appears normal under one detector's representation (e.g., distance-based) may simultaneously violate a different mathematical law (e.g., spectral or dynamical). No single detector spans the full space of mathematical anomaly.

(c) **Lack of attribution:** When an anomaly is detected, practitioners cannot determine whether the deviation is statistical, dynamical, spectral, geometric, or some combination thereof, without running multiple separate analyses.

(d) **Coverage gaps:** High AUC (area under the ROC curve) does not guarantee high coverage—the anomaly scores may concentrate in a narrow band, failing to discriminate among different types of anomalies (the "coverage–AUC gap" problem).

(e) **Hyperparameter sensitivity:** Most detectors require per-dataset tuning of hyperparameters (contamination rate, number of neighbours, kernel bandwidth), limiting practical deployability.

2.2. Prior Art Limitations

Known prior art includes Isolation Forest (Liu et al., 2008), Local Outlier Factor (Breunig et al., 2000), One-Class SVM (Schölkopf et al., 1999), deep autoencoders, out-of-distribution detection methods (ODIN, energy-based methods), and ensemble methods that combine multiple instances of the same detector type. None of these provides multi-law-domain decomposition of anomaly type, per-law attribution, or a formal guarantee that anomalies cannot simultaneously appear normal across all mathematical perspectives.

3. SUMMARY OF THE INVENTION

The present invention provides a method and system for detecting and characterising anomalies by simultaneously evaluating observations under multiple independent mathematical law domains. The key contributions are:

(i) The **Universal Deviation Principle**: a formal mathematical statement, supported by theorems, that if a family of operators $\{\Phi_k\}_{k=1}^{K}$ is *deviation-separating* (i.e., the stacked Jacobian matrix has full rank), then no anomalous observation can simultaneously match normal data across all operators.

(ii) A **multi-spectrum anomaly tensor** $T_{i,k,d}$ constructed by projecting each observation through at least five independent mathematical law domains.

(iii) A **Magnitude–Direction–Novelty (MDN) decomposition** of the anomaly tensor that separates how much, what type, and how novel each deviation is.

(iv) A **Boundary-Centred Dimension Magnifier** that adaptively rescales tensor dimensions based on empirical separability.

(v) A classification stage (QDA-Magnified or Hybrid auto-selection) that produces per-law-domain attributed anomaly scores.

4. DETAILED DESCRIPTION OF THE INVENTION

4.1. Universal Deviation Principle

The method is founded on the following formal principle:

Definition (Deviation-Separating Operator Family): An operator family $\{\Phi_k\}_{k=1}^K$ is *deviation-separating* if the stacked Jacobian matrix:

$$J(x) = \begin{bmatrix} D\Phi_1(x) \\ D\Phi_2(x) \\ \vdots \\ D\Phi_K(x) \end{bmatrix} \tag{1}$$

has full column rank for all x in the data domain.

Theorem 1 (Local Deviation Guarantee): If the operator family $\{\Phi_k\}$ is deviation-separating, then by the inverse function theorem, for any anomalous point x^* not identical to the normal centroid μ_{normal}, there exists at least one operator Φ_k under which x^* deviates from the projection of normal data.

Theorem 2 (Global Deviation Guarantee): With a compact anomaly set, a quantitative lower bound on the minimum deviation across all operators can be established.

Theorem 3 (Finite-Sample Guarantee): Given n normal training samples, the detection guarantee holds with probability at least $1-\delta$ for a sample complexity depending on the operator family's covering number.

Theorem 4 (Information-Optimal Fusion): The optimal weights for combining operator outputs can be computed as the solution to a semidefinite programme (SDP).

Theorem 5 (Greedy Operator Selection): A greedy algorithm for selecting the most informative subset of operators from a large candidate set achieves a $(1 - 1/e)$ approximation to the optimal subset.

4.2. Spectrum Operators (Law Domains)

The method employs at least five fundamental mathematical law domains, each implemented as a spectrum operator Φ_k:

4.2.1 Statistical Operator (Φ_{stat})

Measures deviations in distributional properties:

- Shannon entropy of the observation's local neighbourhood
- Kullback–Leibler divergence from the reference normal distribution
- Hellinger distance from the reference distribution
- Skewness and kurtosis of local feature distributions

4.2.2 Dynamical (Chaos) Operator (Φ_{chaos})

Measures deviations in temporal dynamics:

- Lyapunov exponent estimated from the observation's temporal context
- Recurrence rate from recurrence plot analysis
- Approximate entropy measuring regularity of temporal patterns

4.2.3 Spectral (Frequency) Operator ($\Phi_{\mathbf{freq}}$)

Measures deviations in frequency-domain representations:

- Power spectral density divergence from reference PSD
- Frequency ratio comparing low-frequency to high-frequency energy
- Spectral entropy measuring uniformity of the frequency spectrum
- Spectral centroid shift from the reference centroid

4.2.4 Geometric Operator ($\Phi_{\mathbf{geom}}$)

Measures deviations in geometric relationships:

- Mahalanobis distance from the reference centroid
- Cosine similarity to the reference direction
- Norm ratio comparing observation magnitude to reference magnitude

4.2.5 Exponential Operator ($\Phi_{\mathbf{exp}}$)

Measures deviations under exponential amplification:

$$\Phi_{\text{exp}}(x) = \exp\left(\alpha \cdot \frac{x-\mu}{\sigma}\right) \tag{2}$$

where μ and σ are the reference mean and standard deviation and α is a scaling parameter. This operator amplifies small deviations that may be invisible under linear representations.

4.2.6 Extended Operators

In an enhanced embodiment, additional operators are provided:

- Phase operator: phase-space embedding analysis
- Topological operator: persistent homology features
- Kernel operator: kernel-space deviation
- Rank operator: rank-based deviation statistics
- Wavelet operator: multi-scale wavelet decomposition

In the preferred embodiment, 14 composable operators are available. An operator correlation audit confirms that only 1 of 91 operator pairs exhibits redundancy (Wavelet–Phase, $\rho = 0.92$); all others are sufficiently independent ($\rho < 0.85$).

4.3. Anomaly Tensor Construction

For each observation x_i, each operator Φ_k produces a vector of deviation measurements across D dimensions. The full anomaly tensor is:

$$T_{i,k,d} \quad \text{for observation } i, \text{ operator } k, \text{ dimension } d \tag{3}$$

This tensor captures the multi-law deviation profile of each observation.

4.4. Magnitude–Direction–Novelty (MDN) Decomposition

The anomaly tensor for each observation is decomposed into three interpretable components:

(a) **Magnitude** $r_i = \|T_i\|$: a scalar measuring the total amount of deviation across all operators and dimensions.

(b) **Direction** $\hat{D}_i = T_i/\|T_i\|$: a unit tensor indicating the type of deviation—which operators and dimensions contribute most.

(c) **Novelty** θ_i: the angular distance of $\hat{D}_i$ from reference deviation directions observed in the training data, measuring how novel the deviation pattern is.

This decomposition enables three-axis interpretation: "how much," "what type," and "how novel" the anomaly is.

4.5. Boundary-Centred Dimension Magnifier

A key technical contribution is the Boundary-Centred Dimension Magnifier, which rescales each dimension of the anomaly tensor according to its empirical separability between normal and anomalous classes:

$$r'_d = \tanh\left(k_d \cdot \frac{r_d - b_d}{\sigma_d}\right) \tag{4}$$

where:

- r_d is the raw value in dimension d;
- b_d is the decision boundary location in dimension d;
- σ_d is the scale of the boundary region;
- $k_d = \gamma \cdot \delta_d$ is the steepness, with γ a global parameter and δ_d a per-dimension discriminability score.

The per-dimension discriminability δ_d is computed from three complementary measures:

1. Cohen's d (effect size) between normal and anomalous distributions in dimension d;
2. An overlap metric measuring the overlap area of the two class distributions; and
3. Per-dimension AUC measuring the classification performance using dimension d alone.

In the preferred embodiment, the global parameter $\gamma = 5.0$ and the QDA regularisation parameter $\lambda = 10^{-4}$ are used as a single hyperparameter set across all datasets.

The magnifier has the effect of:

- Amplifying dimensions where normal and anomalous classes are well-separated (high δ_d);
- Suppressing dimensions where the classes overlap heavily (low δ_d);
- Centring the rescaling around the empirical decision boundary rather than the class centroids.

4.6. Classification Stage

The magnified anomaly tensor is classified using one of four strategies, selected automatically via cross-validation (Hybrid auto-selection):

(a) **Fisher projection:** Linear discriminant projection followed by thresholding.

(b) **QDA:** Quadratic discriminant analysis with regularised per-class covariance.

(c) **QDA-Magnified:** QDA applied to the magnifier-rescaled tensor (preferred strategy).

(d) **RankFusion:** A fully unsupervised alternative using rank statistics of per-operator deviations.

The Hybrid auto-selection evaluates all four strategies on cross-validation folds and selects the strategy maximising coverage (the proportion of anomalies detected above threshold) while maintaining AUC above a minimum floor.

4.7. Centroid Estimation

Seven centroid estimation strategies are provided:

1. Sample mean
2. Trimmed mean
3. Geometric median
4. Minimum covariance determinant (MCD)
5. Medoid
6. Huber M-estimator
7. Density-weighted centroid

Auto-selection chooses the centroid strategy that minimises within-class scatter of normal data in the magnified space.

4.8. Coverage Metric

The method introduces a coverage metric that complements AUC. Coverage measures the concentration of anomaly scores in the top-k ranked observations:

$$\text{Coverage@}k = \frac{|\{x : \text{rank}(x) \leq k \text{ and } x \in \text{anomaly}\}|}{|\text{anomaly}|} \tag{5}$$

The coverage metric addresses the AUC–coverage gap: detectors may achieve high AUC while concentrating all anomaly scores in a narrow band, failing to discriminate among different anomaly subtypes. In the preferred embodiment, mean coverage exceeds 91%.

4.9. Performance Characteristics

Using a single hyperparameter set ($\gamma = 5.0$, $\lambda = 10^{-4}$):

Dataset	QDA-Magnified AUC	Hybrid AUC
Synthetic	1.000	0.999
Mimic (camouflage adversarial)	1.000	0.999
Mammography	0.947	0.911
Shuttle	0.989	0.991
Pendigits	0.777	0.960
Mean	**0.943**	**0.972**

5. CLAIMS

The following claims define the scope of protection sought. Where a claim refers to another claim, the dependency is by claim number. Features of any dependent claim may be combined with any independent claim or with other dependent claims.

1. A computer-implemented method for detecting anomalies in data, the method comprising the steps of:

(a) receiving an input observation comprising a feature vector;

(b) projecting the observation through a plurality of spectrum operators, each spectrum operator computing deviation measurements under a different mathematical law domain, the plurality comprising at least:

(i) a statistical operator measuring distributional deviations;
(ii) a dynamical operator measuring temporal dynamical deviations;
(iii) a spectral operator measuring frequency-domain deviations;
(iv) a geometric operator measuring geometric relationship deviations; and
(v) an exponential operator measuring deviations under exponential amplification;

(c) constructing an anomaly tensor from the deviation measurements across all spectrum operators;

(d) decomposing the anomaly tensor into magnitude, direction, and novelty components;

(e) applying a boundary-centred dimension magnifier that rescales each dimension of the anomaly tensor according to an empirically computed per-dimension discriminability score; and

(f) classifying the magnified anomaly tensor to produce an anomaly score and per-law-domain attribution indicating which mathematical law domains contribute to the anomaly determination.

2. A system for detecting and characterising anomalies across multiple mathematical law domains, the system comprising:

(a) a plurality of spectrum operator modules, each configured to compute deviation measurements of an input observation under a different mathematical law domain;

(b) a tensor construction module configured to assemble the deviation measurements into a multi-dimensional anomaly tensor;

(c) a decomposition module configured to decompose the anomaly tensor into magnitude, direction, and novelty components;

(d) a magnifier module configured to rescale each dimension of the anomaly tensor using a boundary-centred rescaling function with per-dimension steepness computed from empirical class separability; and

(e) a classifier module configured to produce an anomaly score and per-law-domain attribution from the magnified tensor;

wherein the plurality of spectrum operators collectively satisfy a deviation-separating condition such that the stacked Jacobian matrix of the operators has full column rank.

3. A method according to claim 1, wherein the boundary-centred dimension magnifier applies the rescaling function:

$$r'_d = \tanh\left(k_d \cdot \frac{r_d - b_d}{\sigma_d}\right)$$

where $k_d = \gamma \cdot \delta_d$, γ is a global steepness parameter, and δ_d is a per-dimension discriminability score computed from Cohen's d, class distribution overlap, and per-dimension AUC.

4. A method according to claim 1, wherein the decomposition into magnitude, direction, and novelty comprises:

(a) computing the magnitude as the norm of the anomaly tensor;

(b) computing the direction as the normalised anomaly tensor; and

(c) computing the novelty as the angular distance of the direction from reference deviation directions in the training data.

5. A method according to claim 1, wherein the classifying step uses quadratic discriminant analysis with regularised per-class covariance matrices applied to the magnified anomaly tensor (QDA-Magnified).

6. A method according to claim 1, further comprising a hybrid auto-selection step that evaluates a plurality of classification strategies on cross-validation folds and selects the strategy maximising a coverage metric while maintaining AUC above a minimum threshold.

7. A method or system according to any preceding claim, further comprising an operator correlation audit step that verifies independence of the spectrum operators by confirming that pairwise correlation coefficients are below a redundancy threshold.

8. A method or system according to any preceding claim, further comprising a centroid auto-selection module configured to select from a plurality of centroid estimation strategies the strategy that minimises within-class scatter of normal data in the magnified space.

9. A method or system according to any preceding claim, wherein a single global hyperparameter set is used across all datasets without per-dataset tuning.

10. A method or system according to any preceding claim, further comprising the step of computing a coverage metric that measures the proportion of anomalies detected in the top-ranked observations, the coverage metric complementing AUC by quantifying score concentration.

11. A method or system according to any preceding claim, wherein the per-law-domain attribution identifies which mathematical law domains—among statistical, dynamical, spectral, geometric, and exponential—contribute most to each anomaly determination, enabling interpretable anomaly diagnosis.

12. A method according to claim 1, further comprising a greedy operator selection step that selects a subset of operators from a larger candidate set, the selection achieving a $(1 - 1/e)$ approximation to the optimal subset under a submodular information criterion.

6. DRAWINGS

[To be filed separately: Diagram of the multi-spectrum operator architecture; illustration of the anomaly tensor construction; visualisation of the MDN decomposition; schematic of the boundary-centred dimension magnifier; operator independence correlation matrix; cross-dataset performance comparison.]

ABSTRACT

A computer-implemented anomaly detection method projects each observation through a plurality of independent mathematical law domains comprising statistical, dynamical, spectral, geometric and exponential operators, constructing a multi-dimensional anomaly tensor. The method is founded on the principle that no anomaly can simultaneously match normal data across all mathematically independent law domains. The anomaly tensor is decomposed into magnitude, direction and novelty components. A boundary-centred dimension magnifier rescales each tensor dimension according to its empirical discriminative power, amplifying well-separated dimensions and suppressing overlapping dimensions. A quadratic discriminant analysis classifier operating on the magnified tensor produces anomaly scores with per-law-domain attribution, enabling interpretable identification of which mathematical aspect renders each observation anomalous. The method achieves a mean ROC-AUC of 0.943 across synthetic, financial, aerospace and medical datasets using a single hyperparameter set.